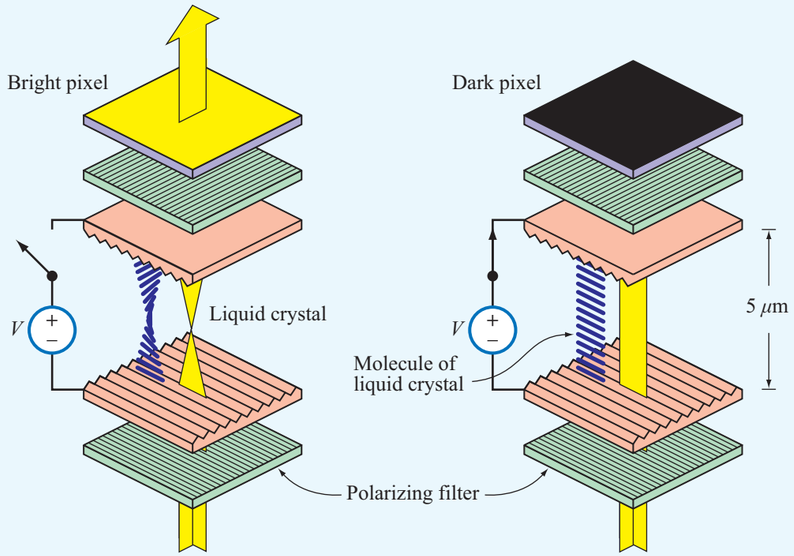


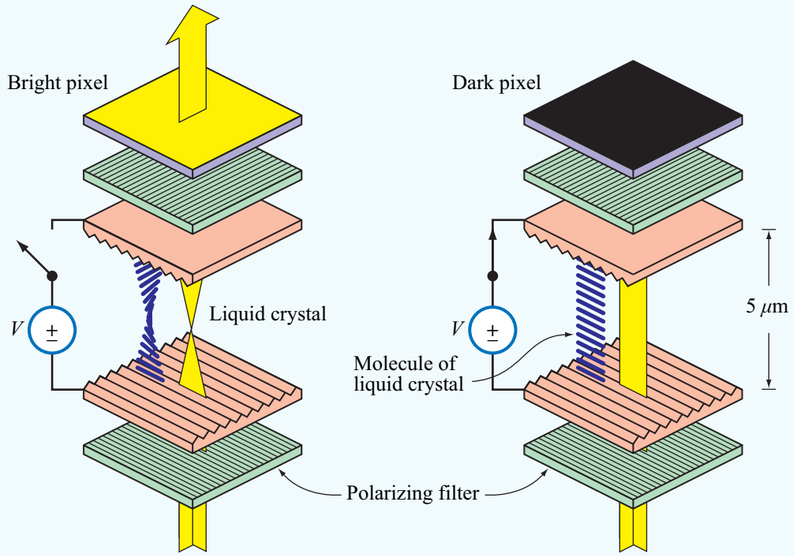
(a) ON state (switch open) (b) OFF state (switch closed)

Figure TF14-2 Single-pixel LCD.

emerges from the exit polarizer, giving the exited surface a ***bright appearance***. A useful feature of nematic liquid crystals is that their spiral untwists (**Fig. TF14-2**(b)) under the influence of an electric field (induced by a voltage difference across the layer). The degree of untwisting depends on the strength of the electric field. With no spiral to rotate the wave polarization as the light travels through the crystal, the light polarization becomes orthogonal to that of the exit polarizer, allowing no light to pass through it. Hence, the pixel exhibits a ***dark appearance***.

► By extending the concept to a ***two-dimensional array*** of pixels and devising a scheme to control the voltage across each pixel individually (usually by using a thin-film transistor), a complete image can be displayed as illustrated in **Fig. TF14-3**. For color displays (**Fig. TF14-4**), each pixel is made up of three subpixels with complementary color filters (red, green, and blue). ◄

7-4 PLANE-WAVE PROPAGATION IN LOSSY MEDIA 321



(a) ON state (switch open) (b) OFF state (switch closed)

图 TF14-2 单像素 LCD。

从出射偏振器中射出，使出射表面具有明亮的外观。向列液晶的一个有用特征是它们的螺旋在电场（由层间电压差引起）的影响下解开（图TF14-2（b））。解绞的程度取决于电场的强度。当光穿过晶体时，由于没有螺旋来旋转波偏振，因此光偏振变得与出射偏振器的偏振正交，从而不允许光通过它。因此，像素呈现出暗的外观。

► 通过将概念扩展到二维像素阵列并设计一种方案来单独控制每个像素上的电压（通常通过使用薄膜晶体管），可以显示完整的图像，如图 TF14-3 所示。对于彩色显示器（图 TF14-4），每个像素由三个具有互补滤色镜（红、绿和蓝）的子像素组成。 ◄

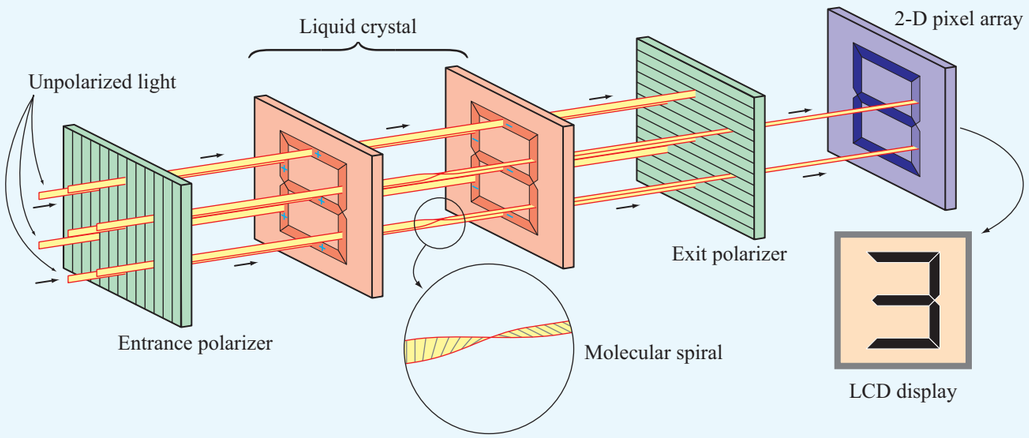


Figure TF14-3 2-D LCD array.



Figure TF14-4 LCD display.

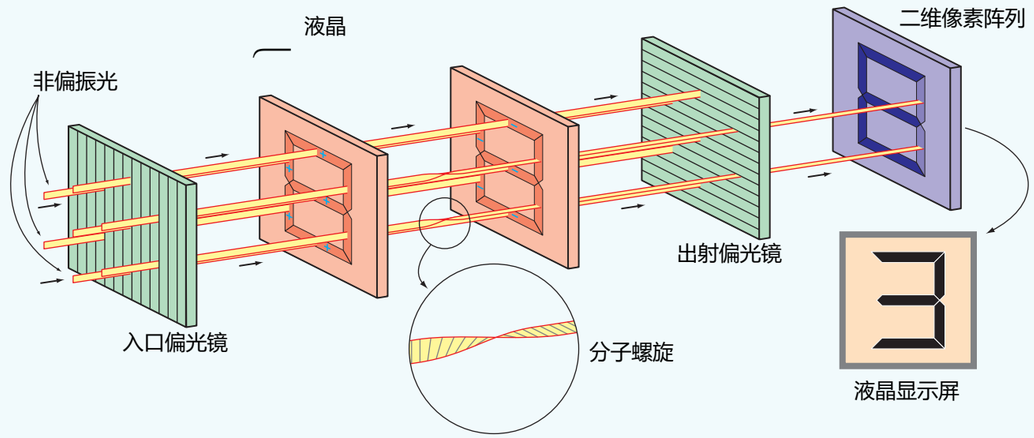


图 TF14-3 二维 LCD 阵列。



图TF14-4液晶显示屏。

7-4.2 Good Conductor

When $\varepsilon''/\varepsilon' > 100$, Eqs. (7.66a), (7.66b), and (7.70) can be approximated as

$$\alpha \approx \omega\sqrt{\frac{\mu\varepsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f\mu\sigma} \quad \text{(Np/m)}, \tag{7.77a}$$

$$\beta \approx \alpha \approx \sqrt{\pi f\mu\sigma} \quad \text{(rad/m)}, \tag{7.77b}$$

$$\eta_c \approx \sqrt{j\frac{\mu}{\varepsilon''}} = (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} = (1+j)\frac{\alpha}{\sigma} \quad (\Omega). \tag{7.77c}$$

(good conductors)

In Eq. (7.77c), we used the relation given by Eq. (1.53): $\sqrt{j} = (1+j)/\sqrt{2}$. For a perfect conductor with $\sigma = \infty$, these expressions yield $\alpha = \beta = \infty$ and $\eta_c = 0$. A perfect conductor is equivalent to a short circuit in a transmission line equivalent.

Expressions for the propagation parameters in various types of media are summarized in Table 7-1.

Example 7-4: Plane Wave in Seawater

A uniform plane wave is traveling in seawater. Assume that the x–y plane resides just below the sea surface and the wave travels in the $+z$ direction into the water. The constitutive parameters of seawater are $\varepsilon_r = 80$, $\mu_r = 1$, and $\sigma = 4$ S/m. If the magnetic field at $z = 0$ is

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,100\cos(2\pi\times10^3 t + 15^\circ) \quad \text{(mA/m)},$$

(a) obtain expressions for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$, and
(b) determine the depth at which the magnitude of $\mathbf{E}$ is 1% of its value at $z = 0$.

Solution: (a) Since $\mathbf{H}$ is along $\hat{\mathbf{y}}$ and the propagation direction is $\hat{\mathbf{z}}$, $\mathbf{E}$ must be along $\hat{\mathbf{x}}$. Hence, the general expressions for the phasor fields are

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}E_{x0}e^{-\alpha z}e^{-j\beta z}, \tag{7.78a}$$

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\,\frac{E_{x0}}{\eta_c}\,e^{-\alpha z}e^{-j\beta z}. \tag{7.78b}$$

To determine α, β, and η_c for seawater, we begin by evaluating the ratio $\varepsilon''/\varepsilon'$. From the argument of the cosine function of $\mathbf{H}(0,t)$, we deduce that $\omega = 2\pi\times10^3$ (rad/s). Therefore, $f = 1$ kHz. Hence,

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{4}{2\pi\times10^3\times80\times(10^{-9}/36\pi)} = 9\times10^5.$$

This qualifies seawater as a good conductor at 1 kHz and allows us to use the good-conductor expressions given in Table 7-1:

$$\alpha = \sqrt{\pi f\mu\sigma} = \sqrt{\pi\times10^3\times4\pi\times10^{-7}\times4} = 0.126 \quad \text{(Np/m)}, \tag{7.79a}$$

$$\beta = \alpha = 0.126 \quad \text{(rad/m)}, \tag{7.79b}$$

$$\eta_c = (1+j)\frac{\alpha}{\sigma} = (\sqrt{2}\,e^{j\pi/4})\frac{0.126}{4} = 0.044e^{j\pi/4} \quad (\Omega). \tag{7.79c}$$

As no explicit information has been given about the electric field amplitude E_{x0}, we should assume it to be complex; that is, $E_{x0} = |E_{x0}|e^{j\phi_0}$. The wave's instantaneous electric and magnetic fields are given by

$$\mathbf{E}(z,t) = \Re\mathfrak{e}\left[\hat{\mathbf{x}}|E_{x0}|e^{j\phi_0}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\right] = \hat{\mathbf{x}}|E_{x0}|e^{-0.126z}\cos(2\pi\times10^3 t - 0.126z + \phi_0) \quad \text{(V/m)}, \tag{7.80a}$$

$$\mathbf{H}(z,t) = \Re\mathfrak{e}\left[\hat{\mathbf{y}}\,\frac{|E_{x0}|e^{j\phi_0}}{0.044e^{j\pi/4}}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\right] = \hat{\mathbf{y}}22.5|E_{x0}|e^{-0.126z}\cos(2\pi\times10^3 t - 0.126z + \phi_0 - 45^\circ) \quad \text{(A/m)}. \tag{7.80b}$$

At $z = 0$,

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,22.5|E_{x0}|\cos(2\pi\times10^3 t + \phi_0 - 45^\circ) \quad \text{(A/m)}. \tag{7.81}$$

By comparing Eq. (7.81) with the expression given in the problem statement,

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,100\cos(2\pi\times10^3 t + 15^\circ) \quad \text{(mA/m)},$$

we deduce that

$$22.5|E_{x0}| = 100\times10^{-3}$$

or

$$|E_{x0}| = 4.44 \quad \text{(mV/m)}$$

and

$$\phi_0 - 45^\circ = 15^\circ \quad \text{or} \quad \phi_0 = 60^\circ.$$

7-4 PLANE-WAVE PROPAGATION IN LOSSY MEDIA 323

7-4.2 良导体

当 $\varepsilon''/\varepsilon' > 100$ 时，等式(7.66a)、(7.66b)和(7.70)可近似为

$$\alpha \approx \omega\sqrt{\frac{\mu\varepsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f\mu\sigma} \quad \text{(Np/m)}, \tag{7.77a}$$

$$\beta \approx \alpha \approx \sqrt{\pi f\mu\sigma} \quad \text{(rad/m)}, \tag{7.77b}$$

$$\eta_c \approx \sqrt{j\frac{\mu}{\varepsilon''}} = (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} = (1+j)\frac{\alpha}{\sigma} \quad (\Omega). \tag{7.77c}$$

(good conductors)

在等式中。(7.77c)，我们使用方程给出的关系。(1.53): $\sqrt{j} = (1+j)/\sqrt{2}$。对于具有 $\sigma = \infty$ 的理想导体，这些表达式产生 $\alpha = \beta = \infty$ 和 $\eta_c = 0$。理想导体相当于传输线等效中的短路。

表7-1总结了各种类型介质中传播参数的表达式。

例7-4：海水中的平面波

均匀的平面波在海水中传播。假设 x–y 平面位于海面正下方，并且波沿 +z 方向传播到水中。海水的本构参数为 $\varepsilon_r = 80$、$\mu_r = 1$和$\sigma = 4$ S/m。如果 z = 0 处的磁场为

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,100\cos(2\pi\times10^3 t + 15^\circ) \quad \text{(mA/m)},$$

(a) 获得 E(z,t) 和 H(z,t) 的表达式，并且
(b) 确定 E 的幅度为其 z = 0 处值的 1% 时的深度。

解：(a) 由于H 沿ŷ 且传播方向为ẑ，因此E 必定沿x̂。因此，相量场的一般表达式为

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}E_{x0}e^{-\alpha z}e^{-j\beta z}, \tag{7.78a}$$

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\,\frac{E_{x0}}{\eta_c}\,e^{-\alpha z}e^{-j\beta z}. \tag{7.78b}$$

为了确定海水的 α、β 和 η_c，我们首先评估比率 $\varepsilon''/\varepsilon'$。根据H(0,t)的余弦函数的参数，我们推导出$\omega = 2\pi\times10^3$ (rad/s)。因此，f = 1 kHz。因此，

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{4}{2\pi\times10^3\times80\times(10^{-9}/36\pi)} = 9\times10^5.$$

这使得海水在 1 kHz 下成为良导体，并允许我们使用表 7-1 中给出的良导体表达式：

$$\alpha = \sqrt{\pi f\mu\sigma} = \sqrt{\pi\times10^3\times4\pi\times10^{-7}\times4} = 0.126 \quad \text{(Np/m)}, \tag{7.79a}$$

$$\beta = \alpha = 0.126 \quad \text{(rad/m)}, \tag{7.79b}$$

$$\eta_c = (1+j)\frac{\alpha}{\sigma} = (\sqrt{2}\,e^{j\pi/4})\frac{0.126}{4} = 0.044e^{j\pi/4} \quad (\Omega). \tag{7.79c}$$

由于没有给出关于电场幅度 E_{x0} 的明确信息，我们应该假设它是复数；即 $E_{x0} = |E_{x0}|e^{j\phi_0}$。波的瞬时电场和磁场由下式给出

$$\mathbf{E}(z,t) = \Re\mathfrak{e}\left[\hat{\mathbf{x}}|E_{x0}|e^{j\phi_0}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\right] = \hat{\mathbf{x}}|E_{x0}|e^{-0.126z}\cos(2\pi\times10^3 t - 0.126z + \phi_0) \quad \text{(V/m)}, \tag{7.80a}$$

$$\mathbf{H}(z,t) = \Re\mathfrak{e}\left[\hat{\mathbf{y}}\,\frac{|E_{x0}|e^{j\phi_0}}{0.044e^{j\pi/4}}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\right] = \hat{\mathbf{y}}22.5|E_{x0}|e^{-0.126z}\cos(2\pi\times10^3 t - 0.126z + \phi_0 - 45^\circ) \quad \text{(A/m)}. \tag{7.80b}$$

At $z = 0$,

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,22.5|E_{x0}|\cos(2\pi\times10^3 t + \phi_0 - 45^\circ) \quad \text{(A/m)}. \tag{7.81}$$

通过比较等式。（7.81）用问题陈述中给出的表达式，

$$\mathbf{H}(0,t) = \hat{\mathbf{y}}\,100\cos(2\pi\times10^3 t + 15^\circ) \quad \text{(mA/m)},$$

我们推断

$$22.5|E_{x0}| = 100\times10^{-3}$$

或者

$$|E_{x0}| = 4.44 \quad \text{(mV/m)}$$

和

$$\phi_0 - 45^\circ = 15^\circ \quad \text{or} \quad \phi_0 = 60^\circ.$$

Table 7-1 Expressions for α, β, η_c, u_p, and λ for various types of media.

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\varepsilon''/\varepsilon' \ll 1$)	Good Conductor ($\varepsilon''/\varepsilon' \gg 1$)	Units
$\alpha =$	$\omega\left[\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}-1\right]\right]^{1/2}$	0	$\approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$	$\approx \sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega\left[\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}+1\right]\right]^{1/2}$	$\omega\sqrt{\mu\varepsilon}$	$\approx \omega\sqrt{\mu\varepsilon}$	$\approx \sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\varepsilon'}}\left(1-j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2}$	$\sqrt{\frac{\mu}{\varepsilon}}$	$\approx \sqrt{\frac{\mu}{\varepsilon}}$	$\approx (1+j)\frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω/β	$1/\sqrt{\mu\varepsilon}$	$\approx 1/\sqrt{\mu\varepsilon}$	$\approx \sqrt{4\pi f/\mu\sigma}$	(m/s)
$\lambda =$	$2\pi/\beta = u_p/f$	u_p/f	$\approx u_p/f$	$\approx u_p/f$	(m)

Notes: $\varepsilon' = \varepsilon$; $\varepsilon'' = \sigma/\omega$; in free space, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon < 0.01$ and a good conducting medium if $\varepsilon''/\varepsilon' > 100$.

Hence, the final expressions for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ are

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,4.44e^{-0.126z}\cos(2\pi \times 10^3 t - 0.126z + 60^\circ) \quad \text{(mV/m)}, \tag{7.82a}$$

$$\mathbf{H}(z,t) = \hat{\mathbf{y}}\,100e^{-0.126z}\cos(2\pi \times 10^3 t - 0.126z + 15^\circ) \quad \text{(mA/m)}. \tag{7.82b}$$

► Because $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ in a lossy medium no longer have the same constant-phase angle, they no longer oscillate in sync with one another as a function of z and t. ◄

(b) The depth at which the amplitude of $\mathbf{E}$ has decreased to 1% of its initial value at $z = 0$ is obtained from

$$0.01 = e^{-0.126z}$$

or

$$z = \frac{\ln(0.01)}{-0.126} = 36.55 \text{ m} \approx 37 \text{ m}.$$

Exercise 7-7: The constitutive parameters of copper are $\mu = \mu_0 = 4\pi \times 10^{-7}$ (H/m), $\varepsilon = \varepsilon_0 \approx (1/36\pi) \times 10^{-9}$ (F/m), and $\sigma = 5.8 \times 10^7$ (S/m). Assuming that these parameters are frequency independent, over what frequency range of the electromagnetic spectrum (see Fig. 1-16) is copper a good conductor?

Answer: $f < 1.04 \times 10^{16}$ Hz, which includes the radio, infrared, visible, and part of the ultraviolet regions of the EM spectrum. (See EM.)

Exercise 7-8: Over what frequency range may dry soil with $\varepsilon_r = 3$, $\mu_r = 1$, and $\sigma = 10^{-4}$ (S/m) be regarded as a low-loss dielectric?

Answer: $f > 60$ MHz. (See EM.)

Exercise 7-9: For a wave traveling in a medium with a skin depth δ_s, what is the amplitude of $\mathbf{E}$ at a distance of $3\delta_s$ compared with its initial value?

Answer: $e^{-3} \approx 0.05$ or 5%. (See EM.)

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表 7-1 各种类型媒体的 α、β、η_c、u_p 和 λ 的表达式。

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\varepsilon''/\varepsilon' \ll 1$)	Good Conductor ($\varepsilon''/\varepsilon' \gg 1$)	Units
$\alpha =$	$\omega\left[\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}-1\right]\right]^{1/2}$	0	$\approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$	$\approx \sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega\left[\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}+1\right]\right]^{1/2}$	$\omega\sqrt{\mu\varepsilon}$	$\approx \omega\sqrt{\mu\varepsilon}$	$\approx \sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\varepsilon'}}\left(1-j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2}$	$\sqrt{\frac{\mu}{\varepsilon}}$	$\approx \sqrt{\frac{\mu}{\varepsilon}}$	$\approx (1+j)\frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω/β	$1/\sqrt{\mu\varepsilon}$	$\approx 1/\sqrt{\mu\varepsilon}$	$\approx \sqrt{4\pi f/\mu\sigma}$	(m/s)
$\lambda =$	$2\pi/\beta = u_p/f$	u_p/f	$\approx u_p/f$	$\approx u_p/f$	(m)

Notes: $\varepsilon' = \varepsilon$; $\varepsilon'' = \sigma/\omega$; in free space, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon < 0.01$ and a good conducting medium if $\varepsilon''/\varepsilon' > 100$.

因此，E(z,t) 和 H(z,t) 的最终表达式为

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,4.44e^{-0.126z}\cos(2\pi \times 10^3 t - 0.126z + 60^\circ) \quad \text{(mV/m)}, \tag{7.82a}$$

$$\mathbf{H}(z,t) = \hat{\mathbf{y}}\,100e^{-0.126z}\cos(2\pi \times 10^3 t - 0.126z + 15^\circ) \quad \text{(mA/m)}. \tag{7.82b}$$

► 由于有损介质中的 E(z,t) 和 H(z,t) 不再具有相同的恒定相位角，因此它们不再作为 z 和 t 的函数彼此同步振荡。 ◄

(b) E 的振幅在 z = 0 处减小至其初始值的 1% 时的深度由下式获得

$$0.01 = e^{-0.126z}$$

或者

$$z = \frac{\ln(0.01)}{-0.126} = 36.55 \text{ m} \approx 37 \text{ m}.$$

练习7-7：铜的本构参数为$\mu = \mu_0 = 4\pi \times 10^{-7}$ (H/m)、$\varepsilon = \varepsilon_0 \approx (1/36\pi) \times 10^{-9}$ (F/m)和$\sigma = 5.8 \times 10^7$ (S/m)。假设这些参数与频率无关，那么在电磁频谱的什么频率范围（见图 1-16）中铜是良导体?

答案：$f < 1.04 \times 10^{16}$ Hz，其中包括电磁频谱的无线电、红外、可见光和部分紫外区域。 （参见EM。）

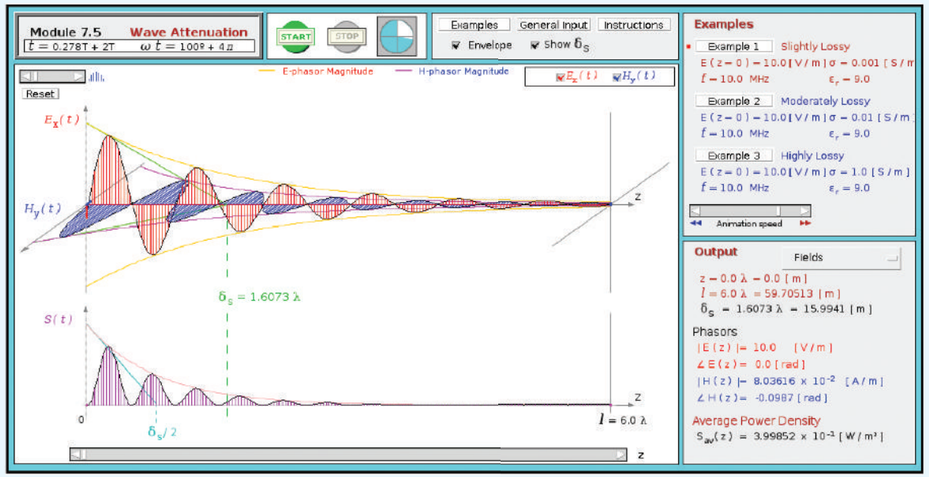
练习 7-8：在什么频率范围内，具有 $\varepsilon_r = 3$、$\mu_r = 1$ 和 $\sigma = 10^{-4}$ (S/m) 的干燥土壤可以被视为低损耗电介质?

Answer: $f > 60$ MHz. (See EM.)

练习7-9：对于在趋肤深度为δ_s的介质中传播的波，与初始值相比，距离$3\delta_s$处E的振幅是多少?

Answer: $e^{-3} \approx 0.05$ or 5%. (See EM.)

Module 7.5 Wave Attenuation Observe the profile of a plane wave propagating in a lossy medium. Determine the skin depth, the propagation parameters, and the intrinsic impedance of the medium.



7-5 Current Flow in a Good Conductor

When a dc voltage source is connected across the ends of a conducting wire, the current flowing through the wire is uniformly distributed over its cross section. That is, the current density **J** is the same along the axis of the wire and along its outer perimeter (**Fig. 7-14(a)**). This is not true in the ac case. As we see shortly, a time-varying current density is maximum along the perimeter of the wire and decreases exponentially as a function of distance toward the axis of the wire (**Fig. 7-14(b)**). In fact, at very high frequencies, most of the current flows in a thin layer near the wire surface, and if the wire material is a perfect conductor, the current flows entirely on the surface of the wire.

Before analyzing a wire with circular cross section, let us consider the simpler geometry of a semi-infinite conducting solid, as shown in **Fig. 7-15(a)**. The solid's planar interface with a perfect dielectric is the x–y plane. If at $z = 0^-$ (just above the surface) an x-polarized electric field with $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}E_0$ exists in the dielectric, a similarly polarized field is induced in the conducting medium and propagates as a plane wave along the $+z$ direction. As a consequence of the boundary condition mandating continuity of the tangential component of **E** across the boundary between any two contiguous media, the electric field at $z = 0^+$ (just below the boundary) is $\widetilde{\mathbf{E}}(0) = \hat{\mathbf{x}}E_0$ also.

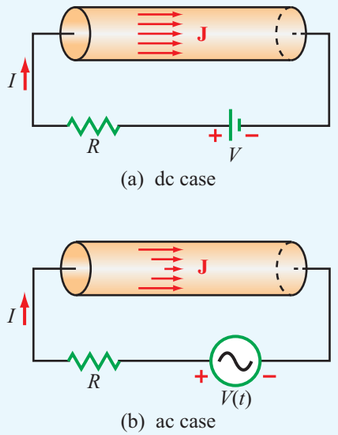
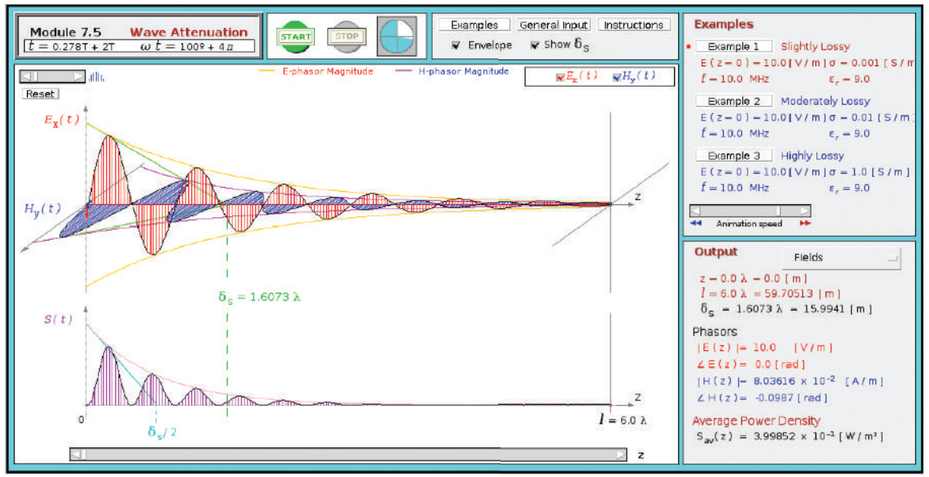


Figure 7-14 Current density **J** in a conducting wire is (a) uniform across its cross section in the dc case, but (b) in the ac case, **J** is highest along the wire's perimeter.

模块 7.5 波衰减 观察有损介质中传播的平面波的轮廓。确定介质的趋肤深度、传播参数和固有阻抗。



7-5 良导体中的电流

当直流电压源连接在导线的两端时，流过导线的电流均匀分布在其横截面上。也就是说，沿着导线的轴线和沿着其外周的电流密度J是相同的（图7-14（a））。但在交流案例中情况并非如此。正如我们很快看到的，随时间变化的电流密度沿着导线的周长最大，并且随着朝向导线轴线的距离而呈指数下降（图7-14（b））。事实上，在非常高的频率下，大部分电流在靠近导线表面的薄层中流动，如果导线材料是理想的导体，则电流完全在导线的表面上流动。

在分析圆形横截面的导线之前，让我们考虑一下半无限导电固体的更简单的几何形状，如图 7-15(a) 所示。固体与理想电介质的平面界面是 x-y 平面。如果在表面) 上方的 $z = 0^-$ (处，电介质中存在具有 $\tilde{E} = \hat{x}E_0$ 的 x 极化电场，则在导电介质中会感应出类似的极化场，并沿 +z 方向作为平面波传播。由于边界条件要求 E 的切向分量在任何两个相邻介质之间的边界上连续，因此边界) 正下方 $z = 0^+$ (处的电场也是 $\tilde{E}(0) = \hat{x}E_0$。

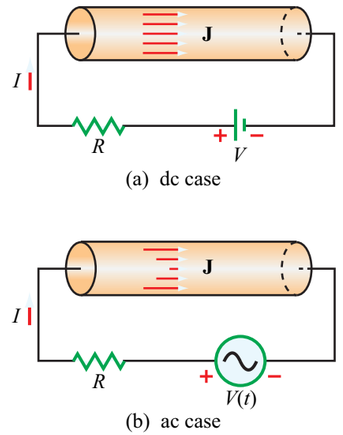


图 7-14 导线中的电流密度 J (a) 在直流情况下在其横截面上是均匀的，但 (b) 在交流情况下，J 沿导线周边最高。